Utkarsh Rai

Robotics Engineer — Robot Learning & Manipulation on Real Hardware
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EDUCATION

Northeastern University — Boston, MA
Master of Science in Robotics — *November 2026*
Relevant Coursework: Robot Mechanics and Control, Mobile Robotics, Optimization, Computer Vision

National Institute of Technology, Silchar — Silchar, India
Bachelor of Technology in Electrical Engineering — *March 2022*

SKILLS

Programming & Systems: C++ (production, real-time), Python, ROS, Linux, Git
Robotics: Kinematics, dynamics, reachability/manipulability/torque analysis, motion planning (RRT*, BIT*), MPC, trajectory optimization
Robot Learning: Behavior cloning, diffusion policies, DAgger, RL (teacher–student distillation, residual RL), sim-to-real transfer
Hardware & Perception: Leader–follower teleoperation, multi-view hand tracking, stereo depth, pose estimation, point clouds
Simulation & Evaluation: Isaac Gym, MuJoCo, Gazebo, regression testing, hardware A/B evaluation

EXPERIENCE

ManiroAI — Robotics Intern — May 2025 – December 2025
Robot learning for contact-rich manipulation — *Boston, MA*

- **Hardware policy deployment:** Distilled a privileged-state RL teacher into a vision–proprioception student deployed on a **7-DoF arm**, retaining **94%** of teacher performance; contributed to the data-collection, deployment, and evaluation workflows.
- **Teleoperation and human-in-the-loop data:** Built a data-collection framework on custom leader–follower hardware, logging **1,000+ hours** of demonstrations, with DAgger-style corrective human feedback to reduce execution drift on hardware.
- **Manipulation evaluation suite:** Built an Isaac Gym CI pipeline across **150+ edge-case scenarios** gated on collision rate and terminal accuracy, catching **32 regressions** before hardware; hardware A/B tests showed task success improving from **78% to 96%**.
- **Multi-view hand tracking:** Developed a pipeline using synchronized cameras and depth-based 3D lifting to generate 3D wrist and joint coordinates for manipulation demonstrations.

Helping Hands Lab, Northeastern University — Graduate Research Assistant — September 2024 – Present
Robot learning and 3D representations for manipulation — *Boston, MA*

- **Sim-to-real for contact-rich insertion:** Developed a bidirectional sim-to-real and real-to-sim adaptation pipeline for point-cloud policies, reducing the performance gap to under **4%** on sub-millimeter insertion.
- **Policies under partial observability:** Trained history-conditioned transformer and recurrent policies, improving task success by **35%** in occluded environments.
- **Point-cloud optimization:** Implemented a factor-graph point-cloud optimization backend with robust covariance estimation and outlier rejection, filtering out **99.4%** of geometric outliers during online multi-sensor mapping.

Mowito — Robotics Engineer — September 2022 – July 2024
Warehouse manipulation — *Bangalore, India*

- **Real-time control on hardware:** Deployed 50 Hz MPC in production C++ on UR5 arms within a **20 ms** per-cycle budget, increasing warehouse pick-and-place throughput by **20%**.
- **Perception-to-control integration:** Integrated a stereo-depth pose-estimation pipeline with the real-time MPC controller, taking it from formulation to live deployment on the robot.
- **Grasp selection:** Ranked grasp candidates in dense, unstructured bins using RGB pose estimation and 3D voxel segmentation.

PUBLICATIONS

- W. Dai, **U. Rai**, J. Chiun, Y. Cao, G. Sartoretti. “Heterogeneous Multi-robot Task Allocation and Scheduling via Reinforcement Learning,” IEEE Robotics and Automation Letters, 10(3):2654–2661, 2025. IEEE Xplore
- **U. Rai**, S. Gangisetty, A. H. Abdul Hafez, A. Subramaniam, C. V. Jawahar. “Visual Place Recognition in Unstructured Driving Environments,” IEEE/RSJ IROS, pp. 10724–10731, 2024. IROS paper
- **U. Rai**, Z. Xu, B.-S. Chen, D. Rosen. “A QCQP-Representable IMU Pre-Integration Factor for Certifiable State Estimation,” arXiv preprint arXiv:2609.38048, 2026. arXiv